

Patterns & Sequences

Discovering the Magic of Patterns Around Us! (Grade 2 Activity Pack)



Future Pattern Detective Notebook



Welcome, Pattern Detective!

Have you ever looked closely at a beautiful **Rangoli** or **Kolam** at your doorstep? Or noticed the gorgeous borders of a **Saree**, the rows of glowing **Diyas** during festivals, or the colorful stack of **Bangles** on your mother's wrist?

Guess what? Those are not just beautiful designs—they are **Mathematical Patterns**! In this book, we are going on a fun adventure to find secrets hidden inside shapes, colors, and numbers. Let's get started!



Our Magical Pattern Chant:

"Pattern dekho, core nikalo, pattern banao!"

(Look at the pattern, find the core, make the pattern!)



What is a Pattern?

A **Pattern** is something that repeats itself over and over again in the exact same order! It follows a special secret rule.

The Secret Key: The Pattern Core!

The **Pattern Core** is the smallest part of the pattern that keeps coming back. If you find the core, you can build or solve any pattern in the world!



Meet the Pattern Families

Patterns love to dress up in different styles! Here are the core types you will see on your mission:

Core Type	What it Looks Like	Real-Life Example	Visual Form
AB	2 alternating elements	Red-Green traffic lights, or two-color bangle stacks.	
AAB	2 of A, then 1 of B	Kantha embroidery: 2 flower stitches then 1 leaf stitch.	
ABB	1 of A, then 2 of B	Beaded festival necklace: 1 red, 2 blue, 1 red, 2 blue...	
ABC	3 different elements cycle	Traffic signals: Red, Yellow, Green cycle.	
AABB	2 of A, then 2 of B	Phulkari border embroidery: 2 red flowers, 2 yellow flowers.	



The 3-Step Strategy to Extend Patterns

When you want to know "What comes next?", follow these 3 golden steps:

Step 1: Find the Core — Circle the very first part that repeats.

Step 2: Count the Repeats — See how many times the core has already appeared.

Step 3: Add the Next Pieces — Put down the next elements in the exact same order!



Growing Patterns & Number Adventures

Some patterns don't just repeat—they **GROW**! Every step they take, they get bigger by the exact same amount. Think of a stack of sweet **Ladoos** growing (1, 2, 3, 4...) or adding ₹5 coins to your piggy bank!



Let's Skip Count!

We can skip count to jump over numbers quickly and find beautiful math paths:

- **Counting by 2s:** 2, 4, 6, 8, 10, 12, 14, 16, 18, 20... (Just like counting cow legs or pairs of shoes!)
- **Counting by 5s:** 5, 10, 15, 20, 25, 30, 35, 40... (Just like counting shiny ₹5 coins!)



Your Pattern Detective Challenge Worksheet

Time to put your detective hat on! Grab your pencils and colors.

Task A: What Comes Next? (Extend the Pattern)

Look at the pattern, find the core, and draw or write the next two pieces!

1. Shape Pattern:



2. Starry Night Pattern:



3. Festival Diya Pattern:

Small Diya, Big Diya, Small Diya, Big Diya, _____, _____

4. Skip Counting by 2s:

2, 4, 6, 8, _____, _____, _____

5. Skip Counting by 5s:

5, 10, 15, 20, _____, _____, _____

Task B: Find the Missing Piece

Oh no! Some pieces ran away. Use the "look before and after" strategy to fill in the blanks.

1. Lost Shape:



2. Color Bead Chain:

Red, Blue, Green, Red, _____, Green, Red, Blue, Green

3. Magical Stars:



4. Number Detective Puzzle (Skip by 3s):

3, 6, 9, _____, 15, 18

Task C: Identify & Name the Pattern Core

Circle the core unit that repeats and write its family name (like AB, AAB, ABC, etc.)

1. Pattern: 🍎 🥤 🍎 🥤 🍎 🥤

Core Unit is: _____ Core Type is: _____

2. Pattern: 🚗 🚗 🚲 🚗 🚗 🚲

Core Unit is: _____ Core Type is: _____

Task D: Create Your Own Masterpiece!

Draw your own beautiful pattern using shapes, colors, or drawings (like smiley faces, flowers, or diamonds). Make it repeat at least 3 times!



1. Draw a circle around your pattern core!
2. What is your Core Type? (Circle one): **AB / AAB / ABB / ABC / AABB**
3. Give your pattern a lovely name! (e.g., "My Sparkly Rangoli Pattern")

Name: _____



Home Exploration & Fun Hunt



The Indian Pattern Hunt Challenge:

Look around your home tonight! Find 2 objects (like a bedsheet, kitchen floor tiles, a plate, or a traditional dress/kurta) that have patterns. Draw them in your rough notebook and tell your family what the **Core** is!

Thank you for learning with PLANRS Academy! Keep your detective eyes open for patterns everywhere!